Problem set 8
Due Tuesday November 10 at 11:59pm

Exercise 8.1 (9 points; Rudin 3.6, modified). Determine whether $\sum_{n=1}^{\infty} a_n$ converges or diverges if

(1) $a_n = \sqrt{n+1} - \sqrt{n}$,

(2) $a_n = \frac{\sqrt{n+1}-\sqrt{n}}{n}$,

(3) $a_n = (n^{1/n} - 1)^n$.

Exercise 8.2 (6 points; Rudin 3.7). Suppose that all $x_n \geq 0$ and $\sum_{n=1}^{\infty} x_n$ converges. Prove that $\sum_{n=1}^{\infty} \frac{\sqrt{x_n}}{n}$ also converges. (Hint: use the Cauchy-Schwarz inequality.)

Exercise 8.3 (10 points). Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} 0 & \text{if } x \notin \mathbb{Q}\,, \\ x^2 & \text{if } x \in \mathbb{Q}\,. \end{cases}$$

Prove that:

(1) $\lim_{x\to 0} f(x) = 0$,

(2) $\lim_{x\to \alpha} f(x)$ does not exist if $\alpha \neq 0$.

Exercise 8.4 (6 points). It is interesting to think about which subsets of $\mathbb{R}$ can arise as the set of points of continuity of a function. For this problem, writing out proofs is optional; if you want, just give examples.

(a) Give an example of a function $f : \mathbb{R} \to \mathbb{R}$ which is continuous at all points in $(-\infty, 0)$, and discontinous at all points in $[0, \infty)$.

(b) Give an example of a function $f : \mathbb{R} \to \mathbb{R}$ which is continuous at all points in $(-\infty, 0]$, and discontinous at all points in $(0, \infty)$.

Exercise 8.5 (not for credit; Rudin 3.16). Fix a positive number α. Choose $x_1 > \sqrt{\alpha}$, and define $x_2, x_3, x_4, \ldots$ by the recursion formula

$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{\alpha}{x_n}\right).$$

(1) Prove that (x_n) is monotonically decreasing and $\lim_{n\to\infty} x_n = \sqrt{\alpha}$.

(2) Define $\varepsilon_n = x_n - \sqrt{\alpha}$. Then show that

$$\varepsilon_{n+1} = \frac{\varepsilon_n^2}{2x_n} < \frac{\varepsilon_n^2}{2\sqrt{\alpha}}$$

and thus, setting $\beta = 2\sqrt{\alpha}$

$$\varepsilon_{n+1} < \beta \left(\frac{\varepsilon_1}{\beta}\right)^{2^n}$$

(3) This is a good algorithm for computing square roots, since the recursion formula is simple and the convergence is extremely rapid. For example, if $\alpha = 3$ and $x_1 = 2$, show that $\varepsilon_1/\beta < \frac{1}{10}$ and therefore $\varepsilon_5 < 4 \cdot 10^{-16}$, $\varepsilon_6 < 4 \cdot 10^{-32}$.